

Evaluating a retrieval pipeline

Experimental setup, primary result, tradeoff, and limitations

The comparison changes one stage at a time

SETUP 02

RESEARCH QUESTION

Can cached indexing cut query latency without reducing Recall@10?

Recall@10

relevant share

Latency p95

end-to-end ms

Query cost

relative compute

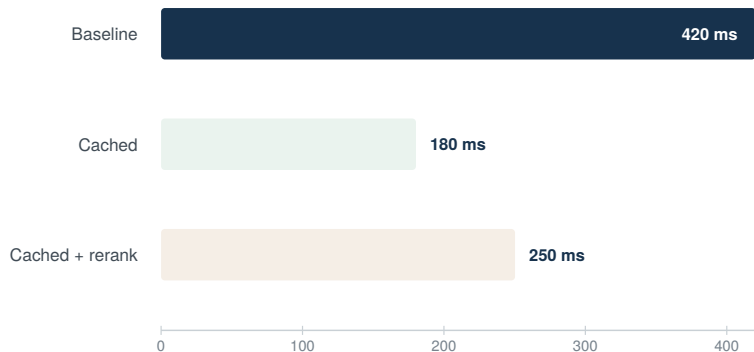
CONTROLLED COMPARISON

Variant	Index	Reranker	Queries	Runs
Baseline	on demand	none	1 000	5
Cached	precomputed	none	1 000	5
Cached + rerank	precomputed	cross-encoder	1 000	5

Cached indexing cuts tail latency

PRIMARY RESULT 03

P95 LATENCY · LOWER IS BETTER



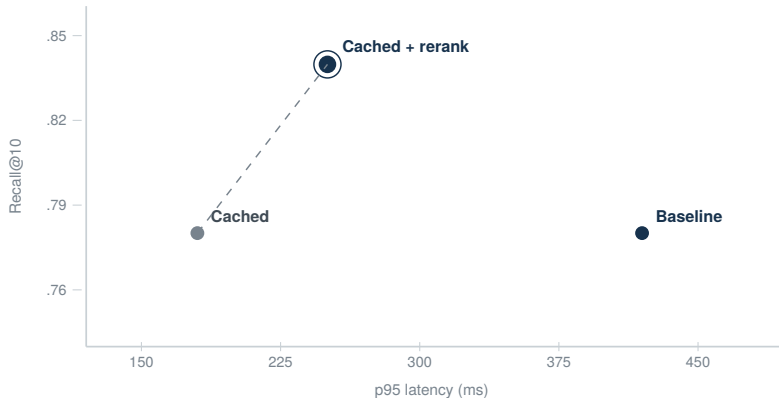
Reranking adds 70 ms

With reranking, p95 remains 40% below baseline.

Uncertainty is omitted from these five-run medians.

Reranking raises recall at 250 ms p95

RECALL@10 VS. P95 LATENCY



Tradeoff

Caching lowers latency while recall remains stable.

Reranking raises recall at a moderate latency cost.

Choose the variant after setting a recall threshold.

Caching lowers latency; evidence is limited

INTERPRETATION

05

SUPPORTED FINDINGS

p95 latency falls from 420 ms to 180 ms.

Recall remains .78 in the illustrative sample.

The reranker raises recall to .84 at 250 ms p95, below the 420 ms baseline.

LIMITS AND NEXT TEST

Five runs cannot establish significance.

Report confidence intervals, corpus-specific outcomes, cache warm-up, and failures.

The next experiment should pre-register the recall threshold and repeat the comparison on unseen queries.

Reporting rule: distinguish observed values, interpretation, and unanswered questions.